



1.2

Rates of Change

Student Notes and Guided Practice

AP Precalculus - Unit 1

| ECHS Mathematics

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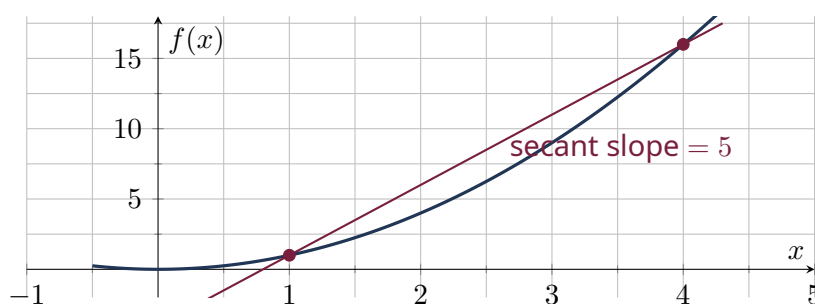
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Learning Targets

- Compute average rate of change from formulas, tables, graphs, and contexts.
- Interpret the sign and units of an average rate of change.
- Compare rates over intervals and use nearby average rates to reason about local behavior.

Essential Question

What does a quotient of changes reveal about how two quantities vary together?



Concept Framework

Key Idea

Average rate of change is the slope of the secant line through two points.

Key Idea

The numerator and denominator must describe changes over the same interval.

Key Idea

Units are output units per input unit.

Key Idea

A positive average rate means the net output change is positive; it does not guarantee the function increased everywhere.

Formula Map**Formula and Structure**

$$\text{AROC}_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

Formula and Structure

$$\text{units} = \frac{\text{output units}}{\text{input units}}$$

Worked Examples**Worked Example 1**

Problem. Find the average rate of change of $f(x) = x^2 - 3x$ on $[1, 5]$.

Solution. $f(5) = 10$, $f(1) = -2$, so $\text{AROC} = (10 - (-2))/(5 - 1) = 3$.

Worked Example 2

Problem. A population changes from 840 to 972 over 6 years. Interpret the average rate.

Solution. $(972 - 840)/6 = 22$ people per year. Over this interval, the population increased by an average of 22 people each year.

Worked Example 3

Problem. A graph has the same value at both interval endpoints. Must its average rate be zero? Must it be constant?

Solution. The average rate is zero because the net change is zero. The function need not be constant between the endpoints.

Multiple-Representation Connection**AP Reasoning**

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Find the average rate of change of $f(x) = 2x^2 - 3x + 1$ on $[1, 5]$.

2. Find the average rate of change of $f(x) = x^2 + 4x - 2$ on $[-2, 3]$.

3. Find the average rate of change of $f(x) = 3x^2 + x$ on $[0, 4]$.

4. Find the average rate of change of $f(x) = -x^2 + 5x + 2$ on $[2, 6]$.

5. Find the average rate of change of $f(x) = x^2 + 1$ on $[1, 4]$.

6. A car travels 42 km in the first 0.6 h and reaches 108 km from its start at 1.5 h. Find and interpret its average rate over $[0.6, 1.5]$.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.

Video support: [Flipped Math topic page](#) | Original ECHS material informed by Pearson and Holt Precalculus concepts.